Question ID: c984f1a5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A hemisphere is half of a sphere. If a hemisphere has a radius of 27 inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

Answer

A. $1{,}500$

B. $6{,}100$

C. $30{,}900$

D. $41{,}200$

Correct Answer: D

Rationale

Choice D is correct. The volume, V, of a sphere is given by $V=\frac{4}{3}\pi r^3$, where r is the radius of the sphere. Since a hemisphere is half of a sphere, it follows that the volume, V, of a hemisphere is given by $V=\left(\frac{1}{2}\right)\left(\frac{4}{3}\right)\pi r^3$, or $V=\frac{2}{3}\pi r^3$. Substituting 27 for r in this formula yields $V=\frac{2}{3}\pi(27)^3$, which gives $V=13{,}122\pi$, or V is approximately equal to $41{,}223.98$. Therefore, the choice that is closest to the volume, in cubic inches, of this hemisphere is $41{,}200$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.